Review 2 Commitment

Form BM04. Completed within 3 working days after Review 2 and confirmed by both reviewers and the supervisor. This document is the sole basis for Defense 1.1 and the Viva. One block per main workflow (section 6) and one block per member (section 7): duplicate or delete blocks to match the team. Export to PDF. File name: SWP490_<ProjectCode>_R2_Commitment.pdf

1. Project Information

Project code	FA26SE238
Project name	DroneOps Intelligence
Team	GFA26SE167
Supervisor	Pham Minh Tri; Dang Ngoc Minh Duc
Review 2 date	27 October 2026, 17:30–20:30
Reviewer 1	PhuongLHK
Reviewer 2	longt5
Version	Review 2 submission v1.0
Date	27 October 2026

2. Final Registration Form

Final version	Pending Review 2 confirmation
Approved at Review 2 (Yes/No)	Pending Review 2 confirmation
Attached file name	SWP490_FA26SE238_R2_Registration.pdf (to be finalized after Review 2)

After this point the project scope does not change.

3. Mandatory Features

ID	Feature	Actor	Acceptance criteria	Workflow	Owner
F-01	Identity, operator and UAV management	Operator; Manager	An authorized user can access only permitted profiles/UAVs; invalid or out-of-scope create/update/read requests are rejected and audited.	WF-01	Ha Quang Nghia
F-02	Mission planning, validation and approval	Operator; Manager	A route is checked at waypoint and segment level; only a current valid request can be approved; rejection leaves current mission content unchanged.	WF-01	Ha Quang Nghia
F-03	MAVLink telemetry and live monitoring	Operator; Manager; Gateway	Authenticated normalized samples are persisted idempotently and visible with source time, freshness, location, altitude, speed, battery and progress within the NFR threshold.	WF-02	Le Quoc Thong
F-04	Operational incident lifecycle	Operator; Manager	Configured abnormal conditions create one traceable incident episode; acknowledgement and resolution retain actor, time, reason and frozen evidence.	WF-02	Le Quoc Thong
F-05	Post-flight history, reports and analytics	Operator; Manager	A completed mission exposes scoped history, route playback, incidents, coverage and calculated metrics; missing evidence is shown as unavailable.	WF-03	Dang Dao Anh Duc

ID	Feature	Actor	Acceptance criteria	Workflow	Owner
F-06	AI-assisted mission assessment	Operator; Manager	Authorized mission context returns traceable advisory output or an explicit unavailable state; provider failure never blocks manual review and AI never approves a mission.	WF-04	Giang Nguyen Gia Khanh

4. Non-Functional Requirements

ID	Quality attribute	Metric	Threshold	Measurement method	Owner
NFR-01	API performance	95th percentile list/detail latency	<= 2 s under P-01	Run >=100 timed requests after warm-up; record hardware/network.	Le Quoc Thong
NFR-02	Realtime performance	Gateway receipt-to-live-view latency	95% <= 2 s	Correlate synchronized gateway-received and client-rendered timestamps.	Le Quoc Thong
NFR-03	Validation performance	Pre-flight validation duration	95% <= 3 s	Run >=30 controlled valid/invalid routes under P-01.	Ha Quang Nghia
NFR-04	Reliability	Monitoring continuity and persisted-record loss	60 min; 0 crash; 0 acknowledged record loss	Controlled soak test plus database and trace comparison.	Le Quoc Thong
NFR-05	Security	Unauthorized/out-of-scope operations rejected	100% of authorization matrix	Positive/negative role and direct-ID API tests.	Le Quoc Thong
NFR-06	Usability	Representative users completing prepared mission task	>=4/5 within 5 min after <=30 min onboarding	Timed task study with a prepared five-waypoint mission.	Le Quoc Thong

5. Out of Scope

Item	Reason
Low-level flight control/autopilot	External to mission-operations software.
Aviation certification/unprovided restrictions	Configured-data validation is not a legal guarantee.
Custom model training	Configured provider integration only.
Native mobile client	Proposed BM03 change; pending Review 2 decision.

6. Workflow Commitments

Minimum number of main workflows = number of team members. Each workflow has exactly one owner. Each mandatory feature belongs to at least one workflow.

6.1 Workflow WF-01

Workflow name	Mission Planning and Pre-flight Validation
Owner (one member)	Ha Quang Nghia
Supporting members	Le Quoc Thong
Actors	Operator; Manager
Trigger	Operator submits a new or changed route.
Business outcome	A reviewed, safe plan is approved or rejected with evidence.
Main entity state path	Mission: Draft -> Pending Review -> Approved/Rejected
Activity diagram (Report 4 section)	Report 4 §3.1, Figure 12

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-01	Identity, operator and UAV management	An authorized user can access only permitted profiles/UAVs; invalid or out-of-scope create/update/read requests are rejected and audited.	TC-F01-01..03
F-02	Mission planning, validation and approval	A route is checked at waypoint and segment level; only a current valid request can be approved; rejection leaves current mission content unchanged.	TC-F02-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-01a	*Main*	*Seeded users/UAV and valid route*	*Create, validate, submit and approve mission*	*Approved current plan with recorded evidence*	*Ha Quang Nghia*
DS-01b	*Exception*	*Approved mission and invalid edit*	*Submit invalid proposal and reject it*	*Old approved content/state remains unchanged*	*Ha Quang Nghia*

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (≥ 3 per feature)	*6*	*4*	*Ha Quang Nghia*
End-to-end workflow test (≥ 1)	*1*	*1*	*Ha Quang Nghia*
State transition tests (≥ 1 per transition)	*4*	*1*	*Ha Quang Nghia*
Total for this workflow	*11*	*6*	*Ha Quang Nghia*

Pass criteria at Defense 1.1

Criterion	Committed
Runs end-to-end on staging with seeded data	*Committed; evidence pending*
Main scenario passes	*Committed; evidence pending*
Exception scenarios passing	*100% committed*

Criterion	Committed
Test pass rate for this workflow	*100% committed*
Open Critical / High defects	*0 required at Defense 1.1*

Reviewer note

Note	Pending Review 2 confirmation.

6.2 Workflow WF-02

Workflow name	Real-Time Monitoring and Incident Management
Owner (one member)	Le Quoc Thong
Supporting members	Ha Quang Nghia
Actors	Gateway; Operator; Manager
Trigger	An approved mission starts and telemetry arrives.
Business outcome	Live state is visible and abnormal events are handled traceably.
Main entity state path	Mission: Approved -> In Progress; Incident: Open -> Acknowledged -> Resolved
Activity diagram (Report 4 section)	Report 4 §3.2, Figures 14–16

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-03	MAVLink telemetry and live monitoring	Authenticated normalized samples are persisted idempotently and visible with source time, freshness, location, altitude, speed, battery and progress within the NFR threshold.	TC-F03-01..03
F-04	Operational incident lifecycle	Configured abnormal conditions create one traceable incident episode; acknowledgement and resolution retain actor, time, reason and frozen evidence.	TC-F04-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-02a	*Main*	*Approved mission and gateway replay*	*Stream samples; raise, acknowledge and resolve incident*	*Fresh live state and immutable incident timeline*	*Le Quoc Thong*
DS-02b	*Exception*	*Active replay with packet loss*	*Stop packets then reconnect*	*One loss episode; recovery is traceable*	*Le Quoc Thong*

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (≥ 3 per feature)	*6*	*4*	*Le Quoc Thong*
End-to-end workflow test (≥ 1)	*1*	*1*	*Le Quoc Thong*
State transition tests (≥ 1 per transition)	*4*	*1*	*Le Quoc Thong*
Total for this workflow	*11*	*6*	*Le Quoc Thong*

Pass criteria at Defense 1.1

Criterion	Committed
Runs end-to-end on staging with seeded data	*Committed; evidence pending*

Criterion	Committed
Main scenario passes	*Committed; evidence pending*
Exception scenarios passing	*100% committed*
Test pass rate for this workflow	*100% committed*
Open Critical / High defects	*0 required at Defense 1.1*

Reviewer note

Note	Pending Review 2 confirmation.

6.3 Workflow WF-03

Workflow name	Post-Flight Analysis and Mission History
Owner (one member)	Dang Dao Anh Duc
Supporting members	Giang Nguyen Gia Khanh
Actors	Gateway; Operator; Manager
Trigger	A trusted completion event is recorded.
Business outcome	Users review route, evidence, incidents, report and analytics.
Main entity state path	Mission: In Progress -> Completed; Report: Queued -> Generating -> Available/Failed
Activity diagram (Report 4 section)	Report 4 §3.3, Figures 17–18

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-05	Post-flight history, reports and analytics	A completed mission exposes scoped history, route playback, incidents, coverage and calculated metrics; missing evidence is shown as unavailable.	TC-F05-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-03a	*Main*	*Completed mission with known evidence*	*Open history, playback, report and analytics*	*Correct scoped metrics, units and coverage*	*Dang Dao Anh Duc*
DS-03b	*Exception*	*Completed mission with telemetry gap*	*Generate and inspect report*	*Gap is unavailable, never silently zero*	*Dang Dao Anh Duc*

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (≥ 3 per feature)	*3*	*2*	*Dang Dao Anh Duc*
End-to-end workflow test (≥ 1)	*1*	*1*	*Dang Dao Anh Duc*
State transition tests (≥ 1 per transition)	*2*	*1*	*Dang Dao Anh Duc*
Total for this workflow	*6*	*3*	*Dang Dao Anh Duc*

Pass criteria at Defense 1.1

Criterion	Committed
Runs end-to-end on staging with seeded data	*Committed; evidence pending*
Main scenario passes	*Committed; evidence pending*
Exception scenarios passing	*100% committed*

Criterion	Committed
Test pass rate for this workflow	*100% committed*
Open Critical / High defects	*0 required at Defense 1.1*

Reviewer note

Note	Pending Review 2 confirmation.

6.4 Workflow WF-04

Workflow name	AI-Assisted Mission Risk Assessment
Owner (one member)	Giang Nguyen Gia Khanh
Supporting members	Dang Dao Anh Duc
Actors	Operator; Manager; AI Provider
Trigger	An authorized user requests an assessment.
Business outcome	Advisory risks and precautions or an explicit failure state support human review.
Main entity state path	Assessment: Pending -> Available/Failed; Mission unchanged
Activity diagram (Report 4 section)	Report 4 §3.4, Figure 19

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-06	AI-assisted mission assessment	Authorized mission context returns traceable advisory output or an explicit unavailable state; provider failure never blocks manual review and AI never approves a mission.	TC-F06-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-04a	*Main*	*Authorized mission and provider stub*	*Request and inspect assessment/provenance*	*Advisory result; no mission-state change*	*Giang Nguyen Gia Khanh*
DS-04b	*Exception*	*Provider timeout fixture*	*Request assessment; continue manual review*	*Explicit unavailable/retry state; review remains usable*	*Giang Nguyen Gia Khanh*

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (≥ 3 per feature)	*3*	*2*	*Giang Nguyen Gia Khanh*
End-to-end workflow test (≥ 1)	*1*	*1*	*Giang Nguyen Gia Khanh*
State transition tests (≥ 1 per transition)	*2*	*1*	*Giang Nguyen Gia Khanh*
Total for this workflow	*6*	*3*	*Giang Nguyen Gia Khanh*

Pass criteria at Defense 1.1

Criterion	Committed
Runs end-to-end on staging with seeded data	*Committed; evidence pending*
Main scenario passes	*Committed; evidence pending*
Exception scenarios passing	*100% committed*

Criterion	Committed
Test pass rate for this workflow	*100% committed*
Open Critical / High defects	*0 required at Defense 1.1*

Reviewer note

Note	Pending Review 2 confirmation.

7. Individual Commitments

One block per member. Every member owns at least one main workflow and 2–3 parts. Each test case and each fix is counted for one member only.

7.1 Member 1

Full name	Le Quoc Thong
Student ID	SE172002
Role in team	Leader; MF2 owner

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	*WF-02*	*Exactly one owner*
Mandatory features owned	*F-03, F-04*	*Primary accountable owner*
Modules / entities owned	*Gateway/telemetry; incidents; live monitoring; frontend/CI and dossier coordination*	*Implementation and evidence*
Key decisions / PRs	*D3; telemetry/incident lifecycle*	*Repository evidence*
Supporting work	*Peer review for another workflow*	*Cross-review distinct from author*

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	*11*	*30 Nov 2026*
Of which automated in CI	*6*	*30 Nov 2026*
Demo scenarios presented	*DS-02a, DS-02b*	*Defense 1.1*
Critical/High defects owned	*0 open at Defense 1.1*	*Defense 1.1*
Documentation/evidence	*Workflow traceability package*	*30 Nov 2026*

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☐ Verified ☐ Partly verified ☐ Not verified	*Pending Review 2 verification*	*Reviewers/team after session*

Member confirmation

Signature / date	

7.2 Member 2

Full name	Ha Quang Nghia
Student ID	SE184490
Role in team	Member; MF1 owner

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	*WF-01*	*Exactly one owner*
Mandatory features owned	*F-01, F-02*	*Primary accountable owner*
Modules / entities owned	*Mission review; validation; schema/state*	*Implementation and evidence*

Type	Items (IDs / names)	Notes
Key decisions / PRs	*D2; mission workflow services*	*Repository evidence*
Supporting work	*Peer review for another workflow*	*Cross-review distinct from author*

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	*11*	*30 Nov 2026*
Of which automated in CI	*6*	*30 Nov 2026*
Demo scenarios presented	*DS-01a, DS-01b*	*Defense 1.1*
Critical/High defects owned	*0 open at Defense 1.1*	*Defense 1.1*
Documentation/evidence	*Workflow traceability package*	*30 Nov 2026*

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☐ Verified ☐ Partly verified ☐ Not verified	*Pending Review 2 verification*	*Reviewers/team after session*

Member confirmation

Signature / date	

7.3 Member 3

Full name	Giang Nguyen Gia Khanh
Student ID	SE182367
Role in team	Member; MF4 owner

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	*WF-04*	*Exactly one owner*
Mandatory features owned	*F-06*	*Primary accountable owner*
Modules / entities owned	*AI context; provider adapter; advisory and failure handling*	*Implementation and evidence*
Key decisions / PRs	*D5; AI assessment evidence to verify*	*Repository evidence*
Supporting work	*Peer review for another workflow*	*Cross-review distinct from author*

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	*6*	*30 Nov 2026*
Of which automated in CI	*3*	*30 Nov 2026*
Demo scenarios presented	*DS-04a, DS-04b*	*Defense 1.1*
Critical/High defects owned	*0 open at Defense 1.1*	*Defense 1.1*
Documentation/evidence	*Workflow traceability package*	*30 Nov 2026*

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☐ Verified ☐ Partly verified ☐ Not verified	*Pending Review 2 verification*	*Reviewers/team after session*

Member confirmation

Signature / date	

7.4 Member 4

Full name	Dang Dao Anh Duc
Student ID	SE194009
Role in team	Member; MF3 owner

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	*WF-03*	*Exactly one owner*
Mandatory features owned	*F-05*	*Primary accountable owner*
Modules / entities owned	*History; reports; analytics/playback*	*Implementation and evidence*
Key decisions / PRs	*D4; post-flight services*	*Repository evidence*
Supporting work	*Peer review for another workflow*	*Cross-review distinct from author*

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	*6*	*30 Nov 2026*
Of which automated in CI	*3*	*30 Nov 2026*
Demo scenarios presented	*DS-03a, DS-03b*	*Defense 1.1*
Critical/High defects owned	*0 open at Defense 1.1*	*Defense 1.1*
Documentation/evidence	*Workflow traceability package*	*30 Nov 2026*

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☐ Verified ☐ Partly verified ☐ Not verified	*Pending Review 2 verification*	*Reviewers/team after session*

Member confirmation

Signature / date	

8. Summary and Cross-check

8.1 By workflow

Workflow	Owner	Features	Demo scenarios (main / exception)	Test cases	Automated
WF-01	*Ha Quang Nghia*	*F-01, F-02*	*1 / 1*	*11*	*6*
WF-02	*Le Quoc Thong*	*F-03, F-04*	*1 / 1*	*11*	*6*
WF-03	*Dang Dao Anh Duc*	*F-05*	*1 / 1*	*6*	*3*
WF-04	*Giang Nguyen Gia Khanh*	*F-06*	*1 / 1*	*6*	*3*

8.2 By member

Member	Workflows owned	Features	Test cases	Automated	Demo scenarios	Review 2 verification
Le Quoc Thong	*WF-02*	*F-03, F-04*	*11*	*6*	*2*	*Pending Review 2*
Ha Quang Nghia	*WF-01*	*F-01, F-02*	*11*	*6*	*2*	*Pending Review 2*
Giang Nguyen Gia Khanh	*WF-04*	*F-06*	*6*	*3*	*2*	*Pending Review 2*
Dang Dao Anh Duc	*WF-03*	*F-05*	*6*	*3*	*2*	*Pending Review 2*

The totals in 8.1 and 8.2 must both equal the committed numbers in 8.3.

8.3 Committed Numbers

Number of main workflows	4
Number of demo scenarios (main / exception)	8 (4 / 4)
Committed functional test cases	34
Of which automated in CI	18 (52.9%)

8.4 Eligibility Thresholds for the Viva

Criterion	Minimum threshold	Agreed threshold
Main workflows	*100% run end-to-end*	*4 (100%)*
Demo scenarios	*100% main; ≥ 80% exception*	*4 main (100%); 4 exception (100%)*
Test case count	*≥ committed number*	*34*
Test pass rate	*≥ 90% overall; 100% main-path tests of mandatory features*	*18 / 34 (52.9%)*
Automation	*≥ 50% run in CI*	*100% per workflow*
Defects	*No open Critical or High defects*	*0*
Individual	*Each member demos the workflow they own and meets their delivery commitments*	*Pending reviewer confirmation*

9. Required Fixes

Items from the Review 2 minutes and Planned items from Form BM02.

ID	Item	Related workflow	Owner	Due
FIX-01	Implement registered Top-k retrieval unless BM03 change is accepted.	WF-04	Giang Nguyen Gia Khanh	30 Nov 2026
FIX-02	Define/remove mobile scope according to BM03 decision.	Cross-cutting	Le Quoc Thong	30 Nov 2026
FIX-03	Align ERD with MissionUpdateRequest, MissionValidationRun and DroneConnection.	WF-01	Ha Quang Nghia	06 Nov 2026
FIX-04	Record seeded staging E2E evidence for all workflows.	WF-01..04	All owners	30 Nov 2026
FIX-05	Canonicalize mission state values and diagrams.	WF-01	Ha Quang Nghia	06 Nov 2026

10. Review 2 Conclusion (completed by reviewers)

☐ Pass ☐ Conditional pass ☐ Re-review

11. Confirmation

Role	Full name	Signature	Date
Reviewer 1	*PhuongLHK*		
Reviewer 2	*longt5*		
Supervisor	*Pham Minh Tri; Dang Ngoc Minh Duc*		
Team representative	*Le Quoc Thong*		

Automatic flight preparation and workflow ownership

Current ownership: MF1 / WF-01 — NghiaHQ (Ha Quang Nghia), planning and preflight; MF2 / WF-02 — ThongLQ (Le Quoc Thong), realtime monitoring, incidents and flight preparation; MF3 / WF-03 — DucDDA (Dang Dao Anh Duc), post-flight; MF4 / WF-04 — KhanhGNG (Giang Nguyen Gia Khanh), AI assessment. ThongLQ remains team leader and dossier coordinator.

Implemented scope

AutomaticFlightRequested is optional and defaults to false. Changes follow the mission proposal and approval workflow. This release prepares readiness; dispatchEnabled is false. Neither confirmation nor gateway enrollment uploads a mission, arms motors, sends TAKEOFF/AUTO/MISSION_START, or changes the mission to InProgress.

Confirmation conditions

For an approved, opted-in mission with a schedule, the backend opens confirmation at T minus 5 minutes and closes it at the scheduled start. The authenticated operator must keep a visible browser tab online. Presence is sent every 10 seconds and expires after 30 seconds. The selected, unrevoked mission connection must match the active UAV and flight-controller UID, with gateway heartbeat within 30 seconds and telemetry within 15 seconds.

Readiness evidence

Confirmation reruns preflight and binds readiness to the exact plan hash and connection ID. Readiness requires a passing preflight for that hash from the last minute. Changed plan/connection, stale presence or missing telemetry invalidates the conditions; the operator must review and confirm again. Dismissal, a hidden/offline tab or a missed window is not flight consent.

Future execution scope

Autopilot execution remains a separate development phase: firmware/airframe capability detection, mission compilation and upload/read-back, command IDs/deadlines and replay protection, scheduled condition rechecks, local pre-arm/failsafe enforcement, and ACK plus observed flight-state verification. Validate ArduPilot/PX4 SITL, propeller-free bench operation and supervised flight before enabling dispatch. AI advice cannot authorize launch.

Additional verification commitment

MF2 owner ThongLQ coordinates readiness window, identity, freshness, plan/connection invalidation, cancellation and no-dispatch verification with MF1 owner NghiaHQ. These are additional planned checks; existing historical test totals/results are not automatically increased or marked executed. Aircraft execution and revised delivery dates require a separate agreed implementation/test baseline.